

The “soler” + INF Periphrasis in Spanish

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1. Introduction

Natural languages employ two main strategies to express frequent, iterated or habitual eventualities: either by means of verbs bearing morphological markers of imperfective aspect (which also allow so-called ongoing and dispositional interpretations), or by means of the dedicated “habitual” particles. In Spanish, rather than a particle, the dedicated construction to express habituality takes the shape of a periphrastic verb involving the verbal predicate *soler* and its infinitive complement.

- (1) a. Juan fuma
Juan smoke.IPFV
‘Juan smokes (as a matter of habit)’
b. Juan suele fumar
Juan SOLER smoke.INF
‘Juan smokes (as a matter of habit)’

The intricacies of the semantic contribution of the dedicated “habitual” or “generic” marker *soler* in Spanish remains largely unexplored, as is the relation it bears to the more general unmarked imperfective mode. In this short paper we present a description of the main syntactic, semantic and pragmatic properties of habitual constructions with *soler*, and sketch a first corresponding formal analysis capturing its semantic properties, its differences with respect to habitual readings of its unmarked imperfective counterpart, and the pragmatic inferences that *soler* is argued to convey.

2. Properties of “soler”

We begin by describing the main syntactic, semantic and pragmatic properties of *soler* periphrases. First and foremost, the main overarching property of *soler* infinitive periphrases is that semantically they always convey a generalizing statement. Among other things, this means that they cannot ever apply to ongoing events, and thus (1b)—unlike its imperfective counterpart (1a)—cannot describe a situation where Juan is currently smoking. When describing ongoing events *soler* statements simply come out false.

2.1. Distribution

From a syntactic standpoint, *soler* shares properties with the distribution of both modal and aspectual periphrases. First, like modal verbs and unlike aspectual periphrases, *soler* cannot appear in the imperative (2):

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- (2) a. *Suele cantar!
 SOLER.IMP sing.INF
 Int.: ‘Have the habit of singing!’
- b. *Puede cantar!
 can.IMP to
 Int.: ‘Have the ability to sing!’
- c. Empieza a cantar!
 begin.IMP to sing.INF
 ‘Start singing!’

However, like aspectual periphrases but unlike modal verbs, *soler* cannot combine with compound infinitives, which typically take perfective morphology ((3); see Gómez Tórrego 1999, Vatrican 2021):

- (3) a. *Suele haberle visto por el Retiro
 SOLER.3SG have.INF-CL see.PFV at the Retiro
 She have see.SOLER him in the Retiro park
- b. Puede haberle visto por el Retiro
 can.IMP have.INF-CL see.PFV at the Retiro
 She could have seen him in the Retiro park

2.2. Iteration

Habitual periphrases with *soler* cannot apply to once-only predicates, as illustrated bellow in (4):

- (4) a. El gorila Maguila vive en libertad
 the gorilla Maguila live.3SG.IPFV in freedom
 ‘Maguilla Gorilla lives freely in the wild’
- b. *El gorila Maguila suele vivir en libertad
 the gorilla Maguila SOLER.3SG live.INF in
 Int.: ‘Maguilla Gorilla live.SOLER freely in the wild’

It seems, thus, that to be felicitous *soler* requires a plurality of certain minimal size, be it a plurality of situations/events as illustrated previously in (1a), or a plurality of individuals, as in (5) below:

- (5) a. Los gorilas viven en libertad
 the gorilas live.3PL.IPFV in freedom
 ‘Gorillas live freely in the wild’
- b. Los gorilas suelen vivir en libertad
 the gorilas SOLER.3PL live.INF in freedom
 ‘Gorillas live.SOLER freely in the wild’

2.3. Actuality

Habitual *soler*-statements may only refer to eventualities that have already been realized. For instance, unlike (6a) below, (6b) cannot describe a situation where the coffee machine is new and was never used before (Green 2000). In such a situation, a statement like (6b) would come out as false.

- (6) a. Esta máquina hace un café muy rico
 this machine make.IPFV a coffee very tasty
 ‘This machine makes very tasty coffee’
 b. Esta máquina suele hacer un café muy rico
 this machine SOLER make.INF a coffee very tasty
 ‘This machine make.SOLER very tasty coffee’

2.4. Exceptions

Generalizations lacking exceptions, including those stating norms, rules, dispositions, uninterrupted states (like e.g. *be a bachelor*), etc. are not describable by means of *soler*, (7).

- (7) a. El sol sale por el este
 the sun come.out-IPFV by the east
 ‘The sun rises in the East’
 b. #El sol suele salir por el este
 the sun SOLER come.out.INF by the east

However, rather than false, such expressions simply result in infelicity or oddness: one could use (7b) to achieve a certain effect, for instance as a joke, within a poetic narrative, etc. Crucially, however, in those instances the normative, dispositional or deontic (circumstantial) flavor of the unmarked *soler*-less counterpart in (7a) is lost.

3. Proposal

In what follows we outline a first formal analysis that aims at capturing the properties of *soler* reported above. We begin first by proposing a baseline analysis for the bare habitual interpretation of imperfective forms in Spanish.

3.1. Bare habitual

Assuming that tense heads are assignment dependent temporal pronouns directly denoting time intervals, I follow much earlier work (Deo 2009, Arregui et al. 2014, Ferreira 2016) in giving habitual interpretations of bare sentences a modal treatment. The modal base is a function from world-time pairs to a set of (ordered) worlds (Kratzer 1981, Kratzer 1991), in this case constituted by a set of gnomic alternatives to world w at time t , where the “dispositions”, “propensities” and “habits” are as in w at t (Boneh & Doron 2008).

In order to compositionally build the intended habitual meaning of Spanish imperfective forms, I assume that in present tense Spanish has a null imperfective morpheme $\emptyset_{\text{IPFV}}$.

$$(8) \quad \llbracket \emptyset_{\text{IPFV}} \rrbracket^{w,t,g} = \lambda P_{\langle s, \langle \epsilon, t \rangle \rangle} . \lambda t' . \forall w' \in MB_w^t . \exists e [*P(w', e) \wedge t' \subseteq T(e)]$$

This morpheme states that every accessible world w' is s.t. the plurality of eventualities $*P$ contains t' , the time of the eventualities at w' . We analyze (1a) as in (9).

$$(9) \quad \llbracket (1a) \rrbracket = \text{T iff } \forall w' \in MB_w^{g(i)} . \exists e [g(i) \subseteq T(e) \wedge \text{smoke}'_{w'}(e) \wedge Ag_{w'}(e) = \text{Juan}]$$

Note that the evaluation world need not be one where e holds, and thus the non-actualization of (6a) is successfully accounted for.

3.2. The building blocks of *soler*

Our account rests on two main pieces. First, we propose that *soler* habituality is a notion that rests on event iteration, as proposed for habituality at large by e.g. Vlach (1993), Boneh & Doron (2008) and Doron & Boneh (2013).¹ Thus, this is first and foremost the key component of *soler*'s semantic contribution: *soler* holds of a plural eventuality e that is equal to the supremum of its proper parts e' (see e.g., Kratzer 2008).

By expressing this requirement via Link's σ -operator (Link 1983), we introduce the necessary existence of a plurality of such e' sub- eventualities:

$$(10) \quad \llbracket \text{V } soler \rrbracket^{w,g,t} = \lambda t'. \exists e [P_w(e) \wedge T(e) \subset t' \wedge e = \sigma e' [P_w(e') \wedge e' < e]]$$

As expressed in (10), *soler* together with perfective aspect simply requires thus iterative eventualities within a topical time, (10).

This reliance on pluralities has two main positive consequences. First, it effectively bans once-only predicates from *soler*-constructions, as desired (see §2.2 above). Moreover, a crucial component of regarding *soler* statements as merely relying on pluralities is that, all else equal, they lack any modal quantification. As a consequence events must be checked with respect to the evaluation world, accounting for *soler*'s restriction to realized eventualities (see §2.3 above).

The second main ingredient of our proposal for *soler* is that not having any quantificational strength of its own, *soler* habituals are quantificationally dependent. As a consequence, they be bound (understood syntactically in terms of c-command) by some quantificational expression, overt or covert (see e.g. Chierchia 1995).

In particular, we follow Cable's (Cable 2022) account of habitual morphology in Tlingit, whereby quantificational adverbs quantify strictly over times in the actual world. The semantic role of such adverbial quantifiers is to determine how often (via quantification) and during what time spans (through their associated contextual restriction) *soler* habitualities occur. For a sentence like (11a), assuming an interpretation of the quantificational expression *every Sunday* as in (11b), we obtain the semantic interpretation in (12):

- (11) a. Suelen ir al fútbol todos los domingos
 SOLER go.INF to soccer every the Sunday
 'They go to a soccer game every Sunday'
- b. $\llbracket \text{every Sunday} \rrbracket^{w,t,g} = \lambda P_{\langle i,t \rangle}. \forall t' [Sunday(t') \wedge t' \in C_t \rightarrow P(t')]$

$$(12) \quad \forall t' [Sunday(t') \wedge t' \in C_t \rightarrow \exists e [T(e) \subset t' \wedge go\text{-}soccer'_w(e) \wedge e = \sigma e' [go\text{-}soccer'_w(e') \wedge e' < e]]]$$

We can see that *soler* + INF habitual periphrases are indeed quantificationally dependent and that they do not introduce quantificational force by themselves by looking into various other quantificational expressions: *soler* statements are compatible with quantificational adverbs of any force, and thus they simply convey iteration over the time frame set by external quantifiers.

- (13) { Siempre / Casi nunca / A veces } suelen ir al fútbol los domingos
 always almost never SOLER go.INF to soccer every the Sunday
 'They {always / almost never / sometimes} go to a soccer game on Sundays (as a matter of habit)'

¹ We do not commit ourselves to reducing habituality at large to event iteration; our claims are in this sense limited to *soler*.

4. Discussion

The proposal sketched above accounts for the fact that *soler* habitual statements (i) are obligatorily habitual, and (ii) convey a plurality of events that (iii) must be actualized in the evaluation world. But nothing so far rules out *soler* habituals in generalizations that are known to have no exceptions (like rules, norms, dispositions, etc.). Our hunch is that such incompatibilities are eminently *pragmatic*, since (i) they lead to oddness as opposed to ungrammaticality and (ii) can be exploited in certain contexts to obtain certain pragmatic effects.

We suggest that *soler* habitual constructions are odd whenever their corresponding bare imperfective counterpart expresses a “rule-based” (in the sense of Carlson 1995) or “definitional” (in the sense of Krifka 2013) generic statement.² Such rule-based generic statements are eminently modal, they uncontroversially convey strong (universal) modal interpretations, and are thus true in all accessible worlds.

- (14)

a.

Bishops move diagonally.

game rules

b.

Tab A fits in slot B.

operating instructions

c.

The Vice-President succeeds the President.

parliamentary rules

Instead, *soler* habituals depend on their associated quantificational adverb to determine the strength of their statement. Even in cases where they combine with adverbs such as *always* (see e.g. (11a) and (13)), these are further contextually restricted, and thus never seldom hold at absolutely all times/situations mandated by the universal quantificational expression at hand in each case.

We suggest that a pragmatically driven strengthening process involving competition between more informative alternatives (for instance along the lines proposed by Magri 2009) may account for this oddness. The price to pay for such kind of solution is that it requires competition between “bare” imperfective forms and *soler* + INF constructions; in other words, the it requires bare habituals to be semantically stronger—i.e. more informative, understood as true in lesser situations/worlds—than *soler* habituals. The advantage of such proposals is that they allow the context to determine whether such stronger alternatives are relevant on a case by case basis with respect to pragmatic competition. For instance, while (15a) may be the default choice of a speaker wanting to describe what’s most usual or the norm, a *soler* statement like (15b) is more apt if the speaker is intending to highlight that, indeed, exceptions to the norm do exist.

- (15)

a.

Los perros tienen cuatro patas
the dogs have.IPFV four legs
‘Dogs have four legs’

b.

Los perros suelen tener cuatro patas
the dogs SOLER.3PL have.INF four legs
‘Dogs (typically) have four legs’

In contrast, *soler* statements that are known to have no counter-examples whatsoever are more difficult to rescue, if possible at all, by means of contextual manipulations of the sort described above, since no exceptions may be highlighted.

- (16)

a.

La tierra da una vuelta alrededor del sol todos los años
the earth give a turn around of-the sun all the years
‘Every year Earth revolves once around the sun’

b.

#La tierra suele dar una vuelta alrededor del sol todos los años
the earth give a turn to-the sun all the years
Int.: ‘(Typically) every year Earth revolves once around the sun’

² These type of generic statement bear some resemblance too to the “in virtue of” generic statements in Greenberg (2003).

- (17) a. En ajedrez los alfiles se mueven en diagonal.
 In chess the bishops REFL move in diagonal
 ‘In chess bishops move diagonally.’
- b. #En ajedrez los alfiles se suelen mover en diagonal.
 In chess bishops REFL SOLER.3PL move.INF in diagonal
 Int.: (Typically) in chess bishops move diagonally.’

If highlighting the presence of an exception to certain rule is what drives the use of a *soler* habitual statement over its bare imperfective counterpart, the contrasts between the pairs in (16) and (17) above follows.

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Proceedings of the 41st West Coast Conference on Formal Linguistics

edited by Nikolas Webster, Yağmur Kiper,
Richard Wang, and Sichen Larry Lyu

Cascadilla Proceedings Project Somerville, MA 2025

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ISBN 978-1-57473-483-6 hardback

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This paper can be cited as:

Mendia, Jon Ander. 2025. The “soler” + INF Periphrasis in Spanish. In *Proceedings of the 41st West Coast Conference on Formal Linguistics*, ed. Nikolas Webster et al., 442-447. Somerville, MA: Cascadilla Proceedings Project. www.lingref.com, document #3776.